Nonlinear Dynamics in Computer Graphics: Reaction–Diffusion Systems in Procedural Generation

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Abstract

[TODO: Abstract goes here.]

1 Introduction

1.1 Objective of this report

This report aims to study the solutions of Reaction Diffusion Systems and explore their far reaching applications outside of mathematics and chemistry. This study will be used in developing a python program that is capable of producing Turing Patterns.

1.2 Structure of this report

The report has been organized to start with a generalist introduction to Reaction Diffusion Systems followed by a mathematical overview of the topic. In particular, we will investigating the solutions of this class of partial differential equations using stability and bifurcation analysis. Studying the solutions of this project is perhaps the most interesting part of this project as they are partly the reason for choosing this topic. The second part of this report will explore the applications of Reaction Diffusion Systems in a variety of domains such as biology, ecology, physics, etc. The final part of this report will demonstrate original contribution from everything that has been learned in the research and conclude with a summary of the topic and the original contribution.

2 Background

2.1 Procedural textures and Perlin noise

Typically, Procedural Generation is done using a technique called *Perlin* Noise developed by Ken Perlin at the Mathematical Animations Groups Inc. (MAGI) while working on the visual effects for the 1982 Disney Sci-fiction film *Tron* Perlin 1985.

The technique uses *interpolation* to geographic terrain and surfaces such as wood and marble. However, Perlin Noise is random and irregular.

There is no way for the artist to precisely control the output to produce organized patterns.

Is there a technique that is capable of producing organized patterns?

2.2 Reaction–diffusion systems

A reaction diffusion system is a *Nonlinear Dynamical system* that models how two or more chemicals *react* and *diffuse* in space and time. A reaction of two chemicals results in new

substances being formed while diffusion occurs when a chemical spreads from a region with lower concentration to higher concentration. [Citation Needed]. General Form of a Reaction Diffusion System:

$$\frac{\partial \mathbf{q}}{\partial t} = D\nabla \mathbf{q} + R(\mathbf{q}), \tag{2.1}$$

$\mathbf{q}(\mathbf{x}, t)$ is the vector of concentrations for each chemical at position $\mathbf{x}$ and time t, $R(\mathbf{q})$ is the reaction component of the sytem , and D is a matrix representing coefficient of diffusions. The Laplacian ∇ compares the concentration of chemicals around a neighbourhood in space and pushes the molecules of a chemical to lower concentration.

Reaction Diffusion Systems can be of multiple components. Components are defined by how many spatial dimensions are used. For instance, a single component RD systems only models the phenonmenon against one space dimension and time. Two component models are needed for producing pattenrns as shown by turing [citaion needed]. There are also many kinds of reaction diffusion system models. In this report, we will be focusing on a simple RD System named the Gray-Scott Model. (Section 2.4).

2.3 Turing patterns

Turing (1952) asked how an embryo, which starts out almost uniform, develops regular structures. His answer was surprising: although diffusion normally smooths things out, if two chemicals diffuse at *different* rates, diffusion can make a uniform state unstable, so that small random disturbances grow into a regular pattern. This is called a *diffusion-driven* or *Turing instability*, and the resulting patterns are called *Turing patterns*. The intuition is short-range activation and long-range inhibition: one substance promotes its own production locally, while the faster-spreading effect of the other suppresses it further away, so peaks form with gaps between them. Reaction–diffusion is now a widely used framework for explaining biological patterns such as animal markings (Kondo and Miura 2010).

The idea entered computer graphics in 1991. Turk (1991) simulated reaction–diffusion directly on the surface of a 3D model, so the texture fits the shape without distortion, and combined several systems to produce leopard rosettes and giraffe patches. Witkin and Kass (1991) extended the method with direction-dependent and spatially varying diffusion. Both show that reaction–diffusion is a practical texture technique, not just a mathematical curiosity.

2.4 The Gray–Scott model

The Gray–Scott model was introduced as a model of an autocatalytic chemical reaction [citaiton needed here]. in a continuously fed, well-mixed tank (Gray and Scott 1984). The chemical rule proposed in the gray scott model is a simplifciation and not an accurate one.

$$U + 2V \to 3V, \qquad V \to P, \tag{2.2}$$

- U reacts 2 molecules of V to produce 3V.
- Therefore, the model is similar to a prey-predator model?
- V also get killed by kill rate.
- U is constantly fed into a well mixed tank.

$$\begin{aligned} \frac{\partial U}{\partial t} &= D_u \nabla^2 U - UV^2 + F(1-U), \\ \frac{\partial V}{\partial t} &= D_v \nabla^2 V + UV^2 - (F+k)V. \end{aligned} \tag{2.3}$$

The term UV^2 is the rate of the first reaction. It is removed from U and added to V, V consumes U to produce another one of itself. The term $F(1-U)$ is the *feed*: how much U is added to the container, pushing U back towards its reservoir value of 1. The equation is dimensionless.

Pearson 1993 did a numerical simulation of the Gray Scott Model and found spatio-irregular patterns.

3 Original Contribution

Stability Analysis

The Gray-Scott system is a nonlinear dynamical system whose equations cannot be solved analytically. Therefore, we consider a method known as stability analysis which will predict the behaviour of the system and precisely identify the set of values for the parameters F and k that lead to the turing patterns we hinted in the previous section. The paramteter values are found using bifurcation analysis

Steps I'll be doing for stability analysis:

- Firstly, I will drop the diffusion terms to show that the system is uniformly stable.
- But before we do that, I solve the system for it's equillibrium points.
- Then classify the system by substituting equillibrium into the jacobian.
- Is it stable or unstable?
- The second part of this analysis will explain what happens when we add back the diffusion term.
- Does the system become unstable? If it does we do a bifurcation analysis to find which parameters F and k will result in turning patterns.

Numerical Simulation of the Gray Scott Equation

This subsection will document my attempt at a numerical simulation of the Gray Model. The numerical simulation is done using a finite difference approximation using 5 point stencil.The code for the simulation is in the appendix. I use the method used by pearson in his paper on Complex patterns in a simple system. I need to complete this section Then I will discuss the results and implications on what i just did.

4 Conclusions

References

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